18 September 1967

SUPERSEDING
MIL-C-10405/2 (SigC)
2 April 1954

MILITARY SPECIFICATION

CRYSTAL UNIT USED IN RADIO RECEIVER BC-499-( )

1. SCOPE

1.1 This specification covers the detail requirements for the crystal unit used with radio receiver BC-499-( ). These crystal units have a quartz vibrator pressure mounted in a plastic holder, type HC-22/U, and are designed to operate at anti-resonance, into a load capacitance of 26 pf, on the fundamental frequency of the vibrator, under noncontrolled temperature conditions, at a specified frequency within the frequency range of 1,875 to 2,875 kilohertz (KHz) , inclusive.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-C-10405 - Crystal Units, Quartz, Pressure and Spacer Mounted.

STANDARDS

MILITARY

MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.

DRAWINGS

MILITARY

SC-DL-12432 - Crystal Holder, HC-22/U, Assembly.
SC-B-22893 - Name Plate, Marking.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 5955

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3. REQUIREMENTS

3.1 Requirements. Requirements shall be in accordance with MIL-C-10405, and as specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection shall be in accordance with MIL-C-10405, and as specified herein.

Table I. Detail requirements.

Requirements	Specified value or condition
Holder	HC-22/U (per drawing SC-DL-12432)
Dimensions	(See SC-DL-12432)
Frequency range	1,875 to 2,875 KHz , incl.
Frequency tolerance	±0.02 percent
Mode of oscillation	Fundamental
Load Capacitance	26 ±0.5 pf
Temperature Range	-40° to + 70°C, incl.
Drive level, rated	(see table II)
Equivalent resistance	220 ohms Maximum
Government reference standard test set	TS-330/TSM
Springs	Minimum force, 2-1/2 lbs., coil type
Marking	Per Drawing SC-B-22893
Shock:	
Frequency change permitted	±0.002 percent or 50 cps, whichever is greater.
	The resistance shall not exceed 220 ohms.
Vibration 1/:	
Frequency change permitted	±0.002 percent or 50 cps, whichever is greater.
	The resistance shall not exceed 220 ohms.
Cleaning test	250 cps (See 3.13 of MIL-C-10405)

1/ Method 201 of MIL-STD-202, amplitude 0.06 inch total excursion.

Table II. Calibration data.

Freq range, incl KHz	Cal. freq., KHz	Crystal Current, ma	Calibrate Resistance, ohms	Drive Level, mw
1,875 to 2,250	2000	20	10	40 ± 10 mw
2,250 to 2,875	2500	30	60	54 ± 10 mw

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5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with MIL-C-10405.

6. NOTES

6.1 The notes specified in MIL-C-10405 are applicable to this specification.

Custodians:
Army - EL

Preparing activity:
Army - EL

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